

Requirements:

1. Implement a trading strategy with at least 3-4 parameters to optimize, such as:
 - A dual moving average crossover strategy with short and long periods
 - RSI thresholds for entry and exit
 - Stop-loss and take-profit percentages
2. Develop three different optimization methods:
 - Grid Search (exhaustive testing of parameter combinations)
 - Random Search (testing random samples from the parameter space)
 - Genetic Algorithm (evolutionary optimization of parameters)
3. Create a robust validation framework that:
 - Splits data into in-sample (training) and out-of-sample (testing) periods
 - Implements time-series k-fold cross-validation
 - Calculates performance consistency across different market regimes
 - Identifies parameter combinations that perform well across all validation sets
4. Implement overfitting detection techniques:
 - Performance degradation analysis (in-sample vs. out-of-sample)
 - Parameter sensitivity testing (how small changes affect performance)
 - Monte Carlo simulation to assess strategy robustness
 - Calculation of statistical significance of results